

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0436

Roanoke Rapids 345 kV
March 16, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0436. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Roanoke Rapids 345 kV substation in the North Carolina area. The request was received on February 16, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 255 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$168,167,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Locks Substation-Deen Point 138kV	Mccartney-North Tower Tap 345kV line	193.6	169	7.4
Floraday-Villa Del Rey 3010 138kV	Mccartney-North Tower Tap 345kV	186.0	289	10.6
Unknown Substation-Walthall 138kV	N Tower Tap-Haggin 345kV line	180.2	317	11.1
N Tower Tap 345 kV/138 kV XFMR	Haggin-North Tower Tap 345kV	178.7	467	16.8
Sonoma 3-Floraday 138kV	Mccartney-N Tower Tap 345kV	177.9	203	7.4
Sonoma 3-Locks 138kV	Mccartney-N Tower Tap 345kV L	177.2	203	7.4
N Tower Tap-Unknown 138kV	N Tower Tap-Haggin 345kV	170.8	344	11.1
Syl Laskin-Elko 138kV	Mitchell Dam-Haggin 345kV line	164.3	200	5.9
Fritch-Big Hanaford 138kV	Mitchell Dam-Skaggs Island Jct 2 Substation 345kV line	163.8	281	8.9
North Tower Tap Substation-Dpc Elroy 138kV L	Mccartney-N Tower Tap 345kV	161.0	286	6.9
Gaston-Racine 138kV Line	N Tower Tap-Mccartney 345kV	156.1	246	7.3
Racine Substation-Lawrenceville 138kV	Mccartney-N Tower Tap Substation 345kV	155.1	239	7.3
Lawrenceville-Skaggs Island 138kV	North Tower Tap-Mccartney 345kV	153.5	237	7.3
Garden-Dpc Elroy 138kV Line	N Tower Tap-Mccartney 345kV L	152.3	223	7.3
Skaggs Island Substation-Garden 138kV	Mccartney-N Tower Tap 345kV	152.3	230	7.3
Dist 000-Birchwood Substation 138kV L	Dunn Siding-Bear Island 345kV line	147.4	304	5.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Roanoke Raids-Gaston 345kV	Mccartney-Roanoke Rapids 345kV	147.2	1,122	79.3
Emporia-Flat Top 138kV L	Mccartney-N Tower Tap 345kV	146.8	232	6.1
Emporia-St. Joseph 138kV	Mccartney-North Tower Tap 345kV	146.5	223	6.1
St. Joseph-Villa Del Rey 3010 138kV	Mccartney-North Tower Tap Substation 345kV	146.5	217	6.1
Hull St Road 345 kV/138 kV Transformer Bank	Mitchell Dam-Skaggs Island Junction 2 345kV line	146.3	529	14.0
Parsons Tap-Flat Top 138kV line	N Tower Tap-Mccartney Substation 345kV	141.4	241	6.1
Trenton Diesel-Dist 1000 138kV	Bear Island-Dunn Siding 345kV line	141.2	325	5.6
Lakeside-Natchez 138kV Line	Mitchell Dam-Skaggs Island Junction 2 345kV Line	140.0	197	6.6
Kewaunee Substation-Goldfield 138kV	Mccartney-N Tower Tap 345kV L	140.0	212	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Locks Substation-Deen Point 138kV	Rebuild line with higher-capacity conductor	\$820,000	36% (\$297,000)
Floraday-Villa Del Rey 3010 138kV L	Rebuild line with higher-capacity conductor	\$11,300,000	42% (\$4,784,000)
Walthall-Uknown 138kV L	Rebuild line with higher-capacity conductor	\$18,264,000	42% (\$7,730,000)
North Tower Tap Substation 345/138 kV XFMR	Replace transformer with higher-rated unit	\$14,535,000	43% (\$6,301,000)
Sonoma 3-Floraday 138kV	Rebuild line with higher-capacity conductor	\$14,390,000	36% (\$5,216,000)
Locks-Sonoma 3 138kV L	Rebuild line with higher-capacity conductor	\$2,712,000	36% (\$983,000)
North Tower Tap-Uknown 138kV	Rebuild line with higher-capacity conductor	\$4,705,000	42% (\$1,991,000)
Syl Laskin-Elko 138kV	Rebuild line with higher-capacity conductor	\$9,388,000	24% (\$2,277,000)
Fritch-Big Hanaford 138kV	Rebuild line with higher-capacity conductor	\$6,569,000	25% (\$1,636,000)
Dpc Eroy-North Tower Tap 138kV	Rebuild line with higher-capacity conductor	\$6,849,000	98% (\$6,720,000)
Racine-Gaston 138kV	Rebuild line with higher-capacity conductor	\$7,283,000	80% (\$5,843,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Racine-Lawrenceville 138kV line	Rebuild line with higher-capacity conductor	\$11,114,000	80% (\$8,917,000)
Lawrenceville-Skaggs Island 138kV	Rebuild line with higher-capacity conductor	\$4,289,000	80% (\$3,441,000)
Garden-Dpc Elroy 138kV Line	Rebuild line with higher-capacity conductor	\$14,665,000	80% (\$11,766,000)
Skaggs Island-Garden 138kV Line	Rebuild line with higher-capacity conductor	\$22,368,000	80% (\$17,946,000)
Dist 1000 Substation-Bircwood 138kV Line	Rebuild line with higher-capacity conductor	\$19,625,000	28% (\$5,485,000)
Roanoke Raids-Gaston 345kV	Rebuild line with higher-capacity conductor	\$37,876,000	80% (\$30,465,000)
Emporia-Flat Top 138kV L	Rebuild line with higher-capacity conductor	\$2,500,000	86% (\$2,153,000)
Emporia-St. Joseph 138kV	Rebuild line with higher-capacity conductor	\$19,646,000	86% (\$16,921,000)
St. Joseph-Villa Del Rey 3010 138kV	Rebuild line with higher-capacity conductor	\$6,985,000	86% (\$6,016,000)
Hull St Road 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$14,017,000	30% (\$4,234,000)
Parsons Tap Substation-Flat Top 138kV	Rebuild line with higher-capacity conductor	\$5,295,000	86% (\$4,561,000)
Trenton Diesel-Dist 1000 138kV	Rebuild line with higher-capacity conductor	\$35,712,000	28% (\$9,981,000)
Lakeside-Natchez 138kV Line	Rebuild line with higher-capacity conductor	\$2,928,000	27% (\$792,000)
Kewaunee Substation-Goldfield 138kV	Rebuild line with higher-capacity conductor	\$4,720,000	36% (\$1,711,000)
	Total allocated to project		\$168,167,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.